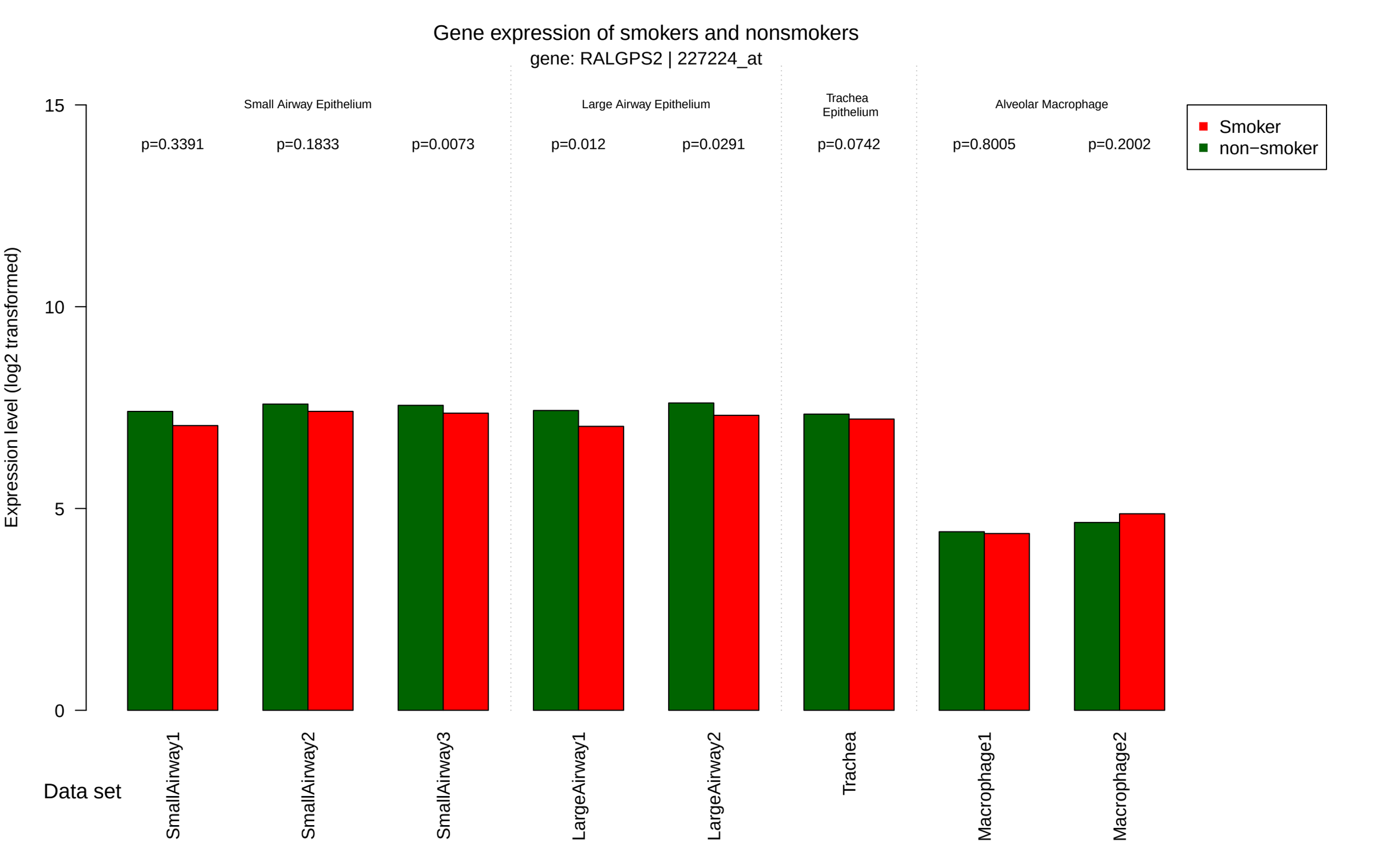
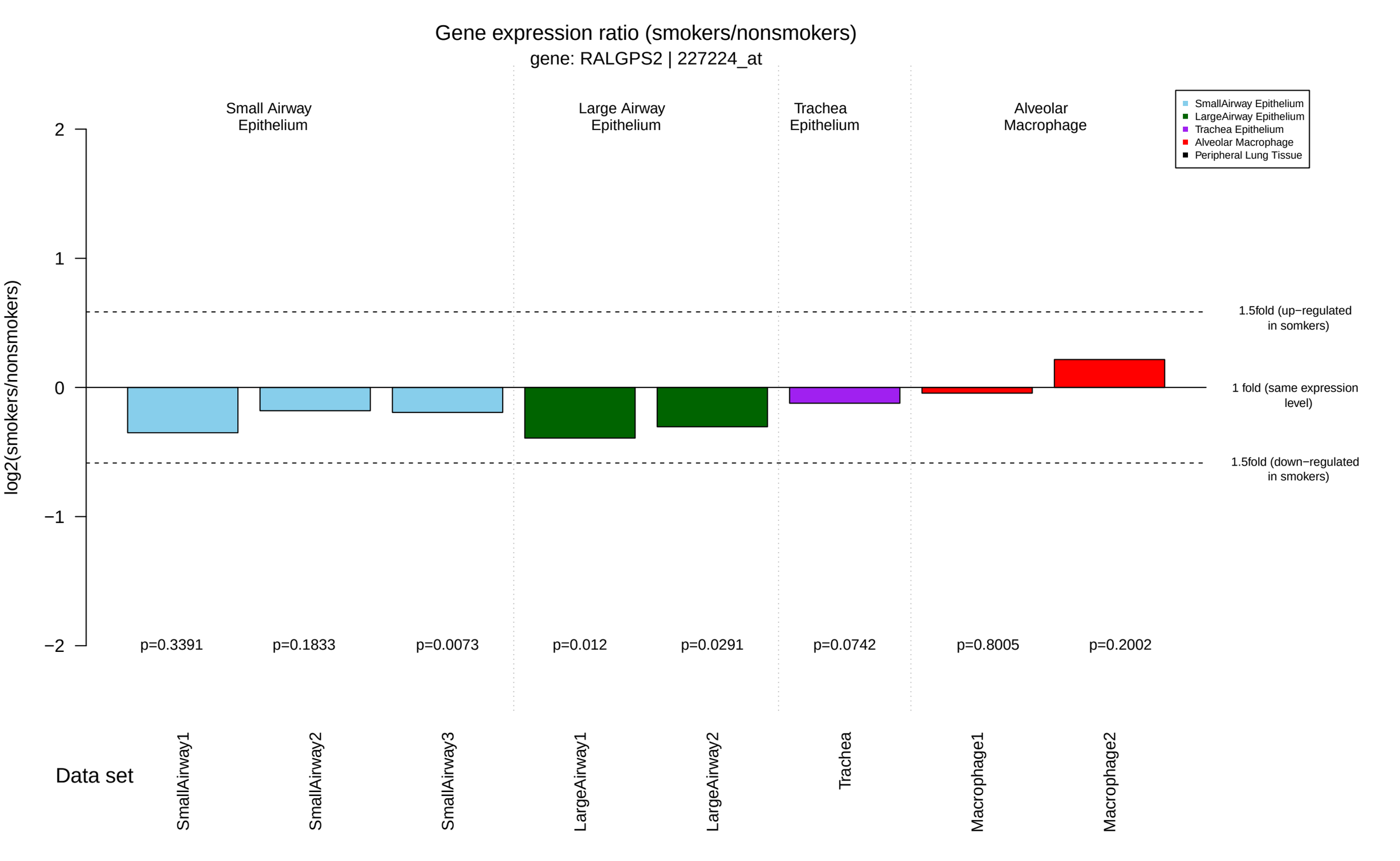
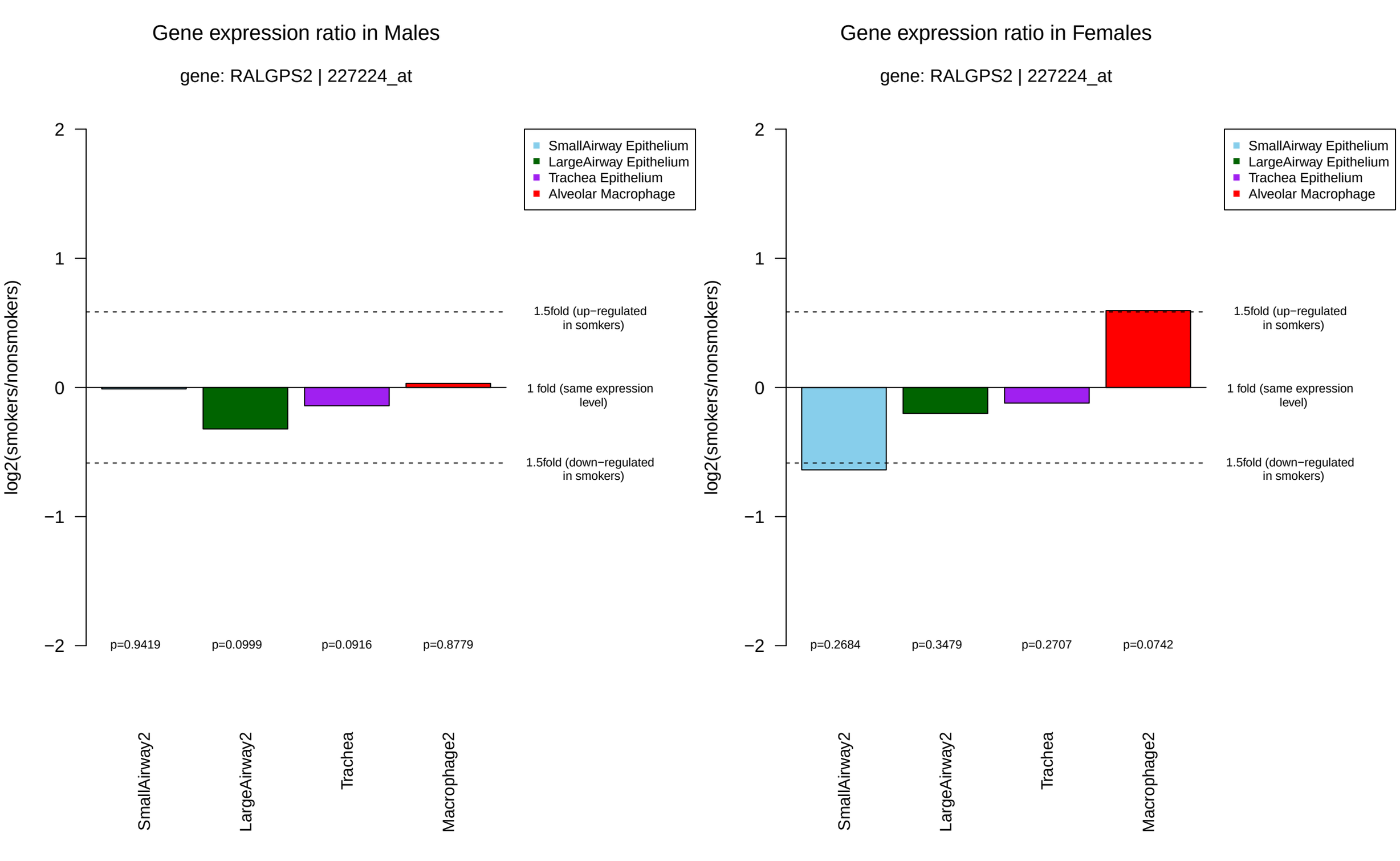
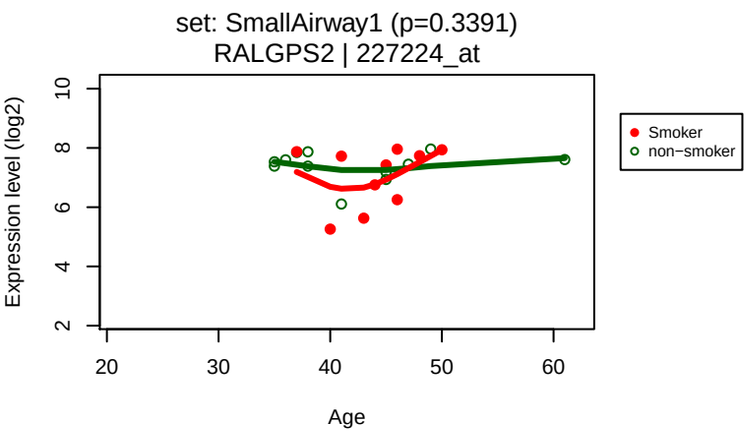
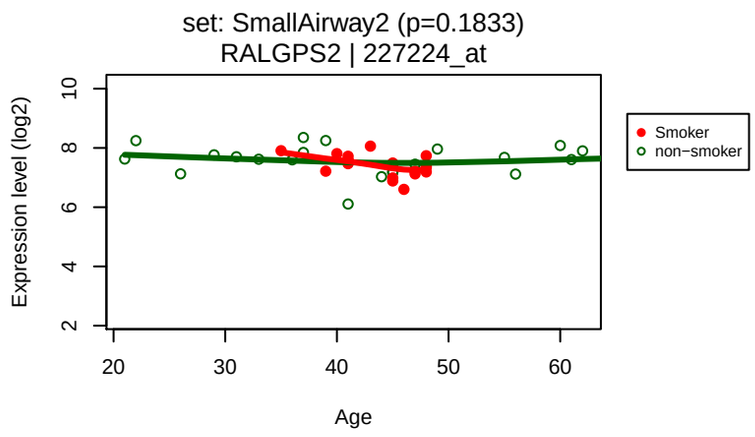
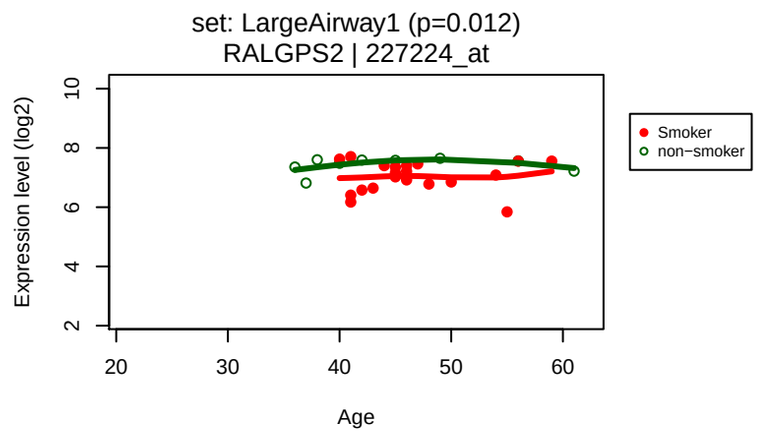

Gene expression of smokers and nonsmokers
gene: RALGPS2 | 227224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3391
p=0.1833
p=0.0073
p=0.012
p=0.0291
p=0.0742
p=0.8005
p=0.2002
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

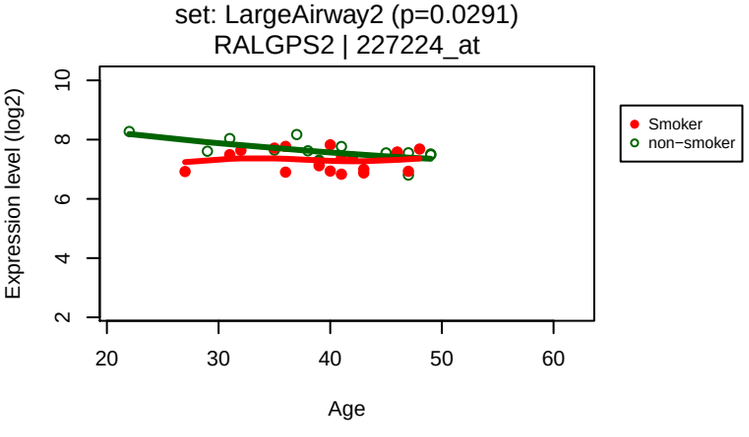

Gene expression ratio (smokers/nonsmokers)
gene: RALGPS2 | 227224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3391
p=0.1833
p=0.0073
p=0.012
p=0.0291
p=0.0742
p=0.8005
p=0.2002
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

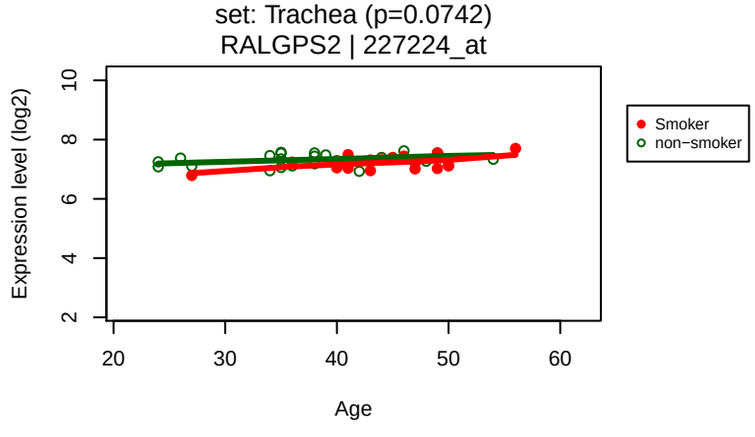

Gene expression ratio in Males
gene: RALGPS2 | 227224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9419
p=0.0999
p=0.0916
p=0.8779
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGPS2 | 227224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2684
p=0.3479
p=0.2707
p=0.0742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

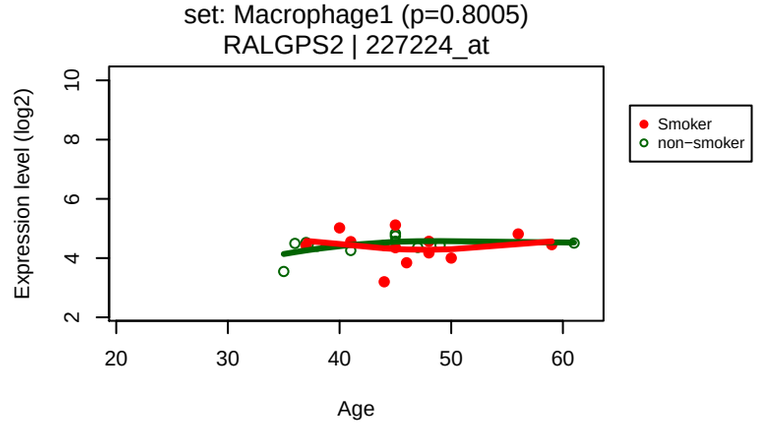

set: SmallAirway1 (p=0.3391)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker

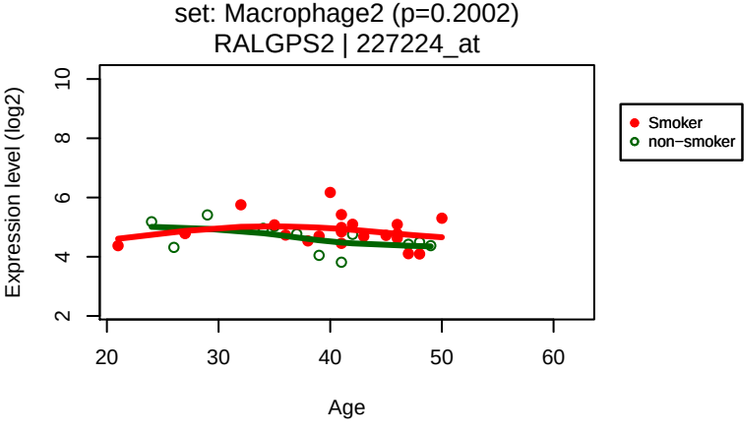

set: SmallAirway2 (p=0.1833)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker

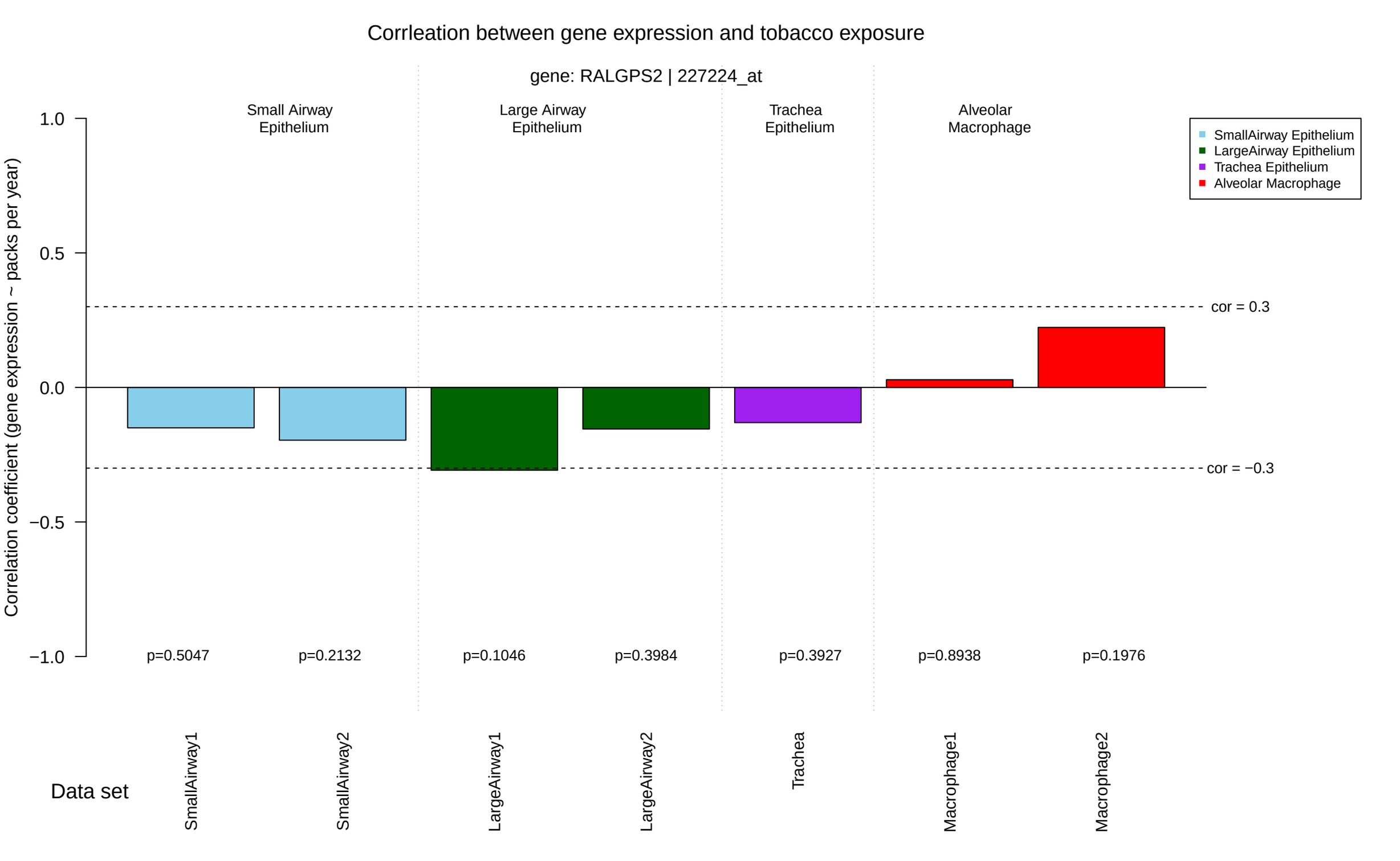

set: LargeAirway1 (p=0.012)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0291)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0742)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8005)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2002)
RALGPS2 | 227224_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGPS2 | 227224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5047
p=0.2132
p=0.1046
p=0.3984
p=0.3927
p=0.8938
p=0.1976
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2